MILITARY SPECIFICATION

AMPLIFIER, ANALOG VOLTAGE: 10554887

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers analog voltage amplifier 10554887 used in power supply 10554531, part of Radar Test Set AN/TPM-22. The analog voltage amplifier provides reference voltages for test equipment in Fire Control Test Set AN/TPM-22.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction; M18; Manufacturing Process and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

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DRAWINGS

U. S. Army, Frankford Arsenal

D10554887	Amplifier, Analog Voltage

PACKAGING DATA SHEET

10554887	Packaging of Analog Voltage Amplifier

(Copies of specification, standards, drawings, and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The analog voltage amplifier shall be manufactured in accordance with Drawing D10554887 and drawings pertaining thereto.

3.2 General specification.- Where applicable, the analog voltage amplifier shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration.- The analog voltage amplifier shall be mounted on the vibration table in normal service position. The analog voltage amplifier shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range	Double Amplitude
5 to 15 Hz	0.06 inch
15 to 25 Hz	0.04 inch
25 to 55 Hz	0.02 inch

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The rate of change of frequency at each frequency range shall not exceed one cycle-per-second per second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock.- The analog voltage amplifier shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 $\pm$ 1 milliseconds. The G value of each impulse shall be within $\pm$ 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds.

3.3.3 Altitude.

3.3.3.1 Operating.- The analog voltage amplifier shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating.- The analog voltage amplifier shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating.- The analog voltage amplifier shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Nonoperating (storage).- The analog voltage amplifier shall not suffer permanent damage when subjected to the temperature extremes of -80°F and +185°F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity-nonoperating.- The analog voltage amplifier shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.- Each conductor shall be electrically continuous as specified on the schematic diagram.

3.5 Performance.- With power applied to analog voltage amplifier in accordance with Table I, analog voltage amplifier shall yield outputs described below.

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TABLE I - ANALOG VOLTAGE AMPLIFIER INPUTS

VOLTAGE APPLIED*	P1 TERMINALS
+15V	36 & 37
+15V RET.	22 & 23
-15V	20 & 21
-15V RET.	34 & 35
+23V	55 & 56
+23V RET.	51 & 52
-23V	49 & 50
-23V RET.	53 & 54

*All voltages applied shall be regulated to ±1%, capable of up to 20 milliamperes, with peak-to-peak ripple not greater than 3 millivolts.

3.5.1 -10 volts dc output.- Analog voltage amplifier shall yield -10.0 ±0.010 volts dc between TP1 (-10) and TP4 (COM). The Output Voltage will be adjusted by R5 to the level stated above into a 500 ohm ±1% load.

3.5.2 +10 volts dc output.- Analog voltage amplifier shall yield +10.0 ±0.010 volts dc between TP2 (+10) and TP4 (COM) into a 500 ohm ±1% load.

3.5.3 +20 volts dc output.- Analog voltage amplifier shall yield +20.0 ±0.020 volts dc between TP3 (+20) and TP4 (COM) into a 2K ohm ±1% load.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of one analog voltage amplifier manufactured in the same manner, using the same materials, equipment, processes and procedures as used

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in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot information, and presentation of lots shall be in accordance with Submission of Product provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein. Examinations and test for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

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TABLE II - CLASSIFICATION OF DEFECTS

100% INSPECTION

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101. Electrical	3.4	4.6.2.1
102. -10 volts dc output	3.5.1	4.6.3.1
103. +10 volts dc output	3.5.2	4.6.3.2
104. -20 volts dc output	3.5.3	4.6.3.3

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- One analog voltage amplifier out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table III.

TABLE III - SPECIAL SAMPLING

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable dwg. visual
302. General specification	3.2	MIL-F-13926 visual
303. Vibration	3.3.1	4.6.1.1
304. Shock	3.3.2	4.6.1.2
305. Altitude - operating	3.3.3.1	4.6.1.3
306. Altitude - nonoperating	3.3.3.2	4.6.1.3
307. Temperature	3.3.4	4.6.1.4
308. Humidity - nonoperating	3.3.5	4.6.1.5

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

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4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment.- Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Contractor design.- The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is considered normally to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to ensure that the test equipment will indicate reliable acceptability of a product that does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

4.5.2.2 Test equipment.- In conjunction with 4.5.2.1, the following test equipment shall be used in the performance of the applicable tests as specified in 4.6:

ITEM NUMBER	DESCRIPTION
1. Power supply	+15 volts dc. regulation ±1%, 20 milliamperes, peak-to-peak ripple not greater than 3 millivolts.
2. Power supply	-15 volts dc, regulation ±1%, 20 milliamperes, peak-to-peak ripple not greater than 3 millivolts.
3. Power supply	+23 volts dc, regulation ±1%, 20 milliamperes, peak-to-peak ripple not greater than 3 millivolts.
4. Power supply	-23 volts dc regulation ±1%, 20 milliamperes, peak-to-peak ripple not greater than 3 millivolts.
5. Digital voltmeter	Dc range at least 0.001 to 50 volts. Accuracy at least 0.2% of full scale.

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4.6 Test methods and procedures.

4.6.1 Environmental.- Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration.- The test shall be performed as specified in Method 514, Procedure IX, Part I, of MIL-STD-810 except that Failure Criteria of 4.9.1.3 does not apply. After vibration, the analog voltage amplifier shall be tested as given in the subparagraphs of 4.6.2 and 4.6.3 for compliance with 3.3.1.

4.6.1.2 Shock.- This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be 11 $\pm$ 1 ms and 15 g's. After the shock impulses, the analog voltage amplifier shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

4.6.1.3 Altitude non-operating and operating.- This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in. Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and non-operating).- This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-810. The temperature of Step 2 shall be +185°F. The temperature of Step 4 shall be +130°F. In Step 5 the analog voltage amplifier shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.4.2 Low (operating and non-operating).- This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -85°F maintained for 24 hours. The temperature of Step 4 shall be +32°F. In Steps 5 and 7, the analog voltage amplifier shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity-nonoperating.- This test shall be performed in accordance with Procedure I, Method 507 of MIL-STD-810 except that in Step 4, the relative humidity shall be maintained at 95%.

4.6.2 Electrical.- Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance.- Before performing test procedures, connect power supplies, items 1 through 4 as specified in Table I. (See Figure 1).

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4.6.3.1 -10 volts dc output.- Using digital voltmeter, item 5, measure voltage between TP1 (-10) and TP4 (COM). Voltage shall be +10.0 ±0.010 volts dc.

4.6.3.2 +10 volts dc output.- Using digital voltmeter, item 5, measure voltage between TP2 (+10) and TP4 (COM). Voltage shall be +10.0 ±0.010 volts dc.

4.6.3.3 +20 volts dc output.- Using digital voltmeter, item 5, measure voltage between TP3 (+20) and TP4 (COM). Voltage shall be +20.0 ±0.010 volts dc.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10554887. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use.- Analog voltage amplifier is used in power supply 10554531, part of Radar Test Set AN/TPM-22. Analog voltage amplifier provides dc reference voltages of -10, +10, and +20 for test equipment in Fire Control Test Set AN/TPM-22.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable packaging data sheet (see 5.1).
c. Selection of applicable level of preservation packaging and packing.
d. Level of protection required.
e. Applicable stock number.
f. Block of serial numbers to identify each analog voltage amplifier.

Custodian:

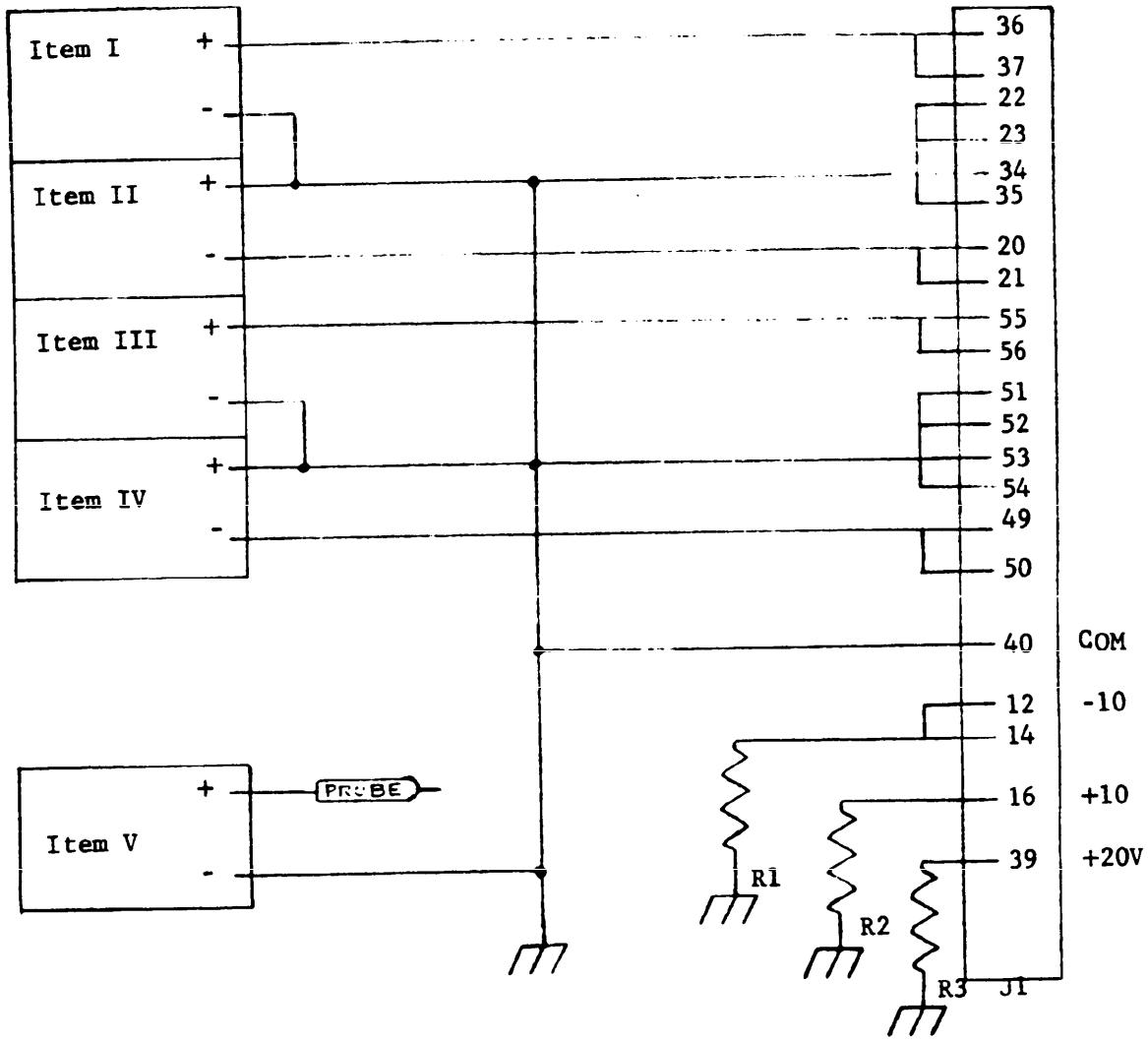
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R1 and R2 = 500 ohm ±1% 1 watt.
R3 = 2K ohm ±1% 1 watt.
J1 = Connector, 5893232, Malco Mfg. Company, Inc., Chicago, IL or equal

FIGURE 1 TEST FIXTURE

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DOCUMENT IDENTIFIER AND TITLE
MIL-A-50300A(MU) Amplifier, Analog Voltage: 10554887

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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